LEARN MATES

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Note from author

These notes are mostly based on the syllabus and past papers

Fair warning: they might look short (2–3 pages), but I've condensed everything down to keep the important stuff while removing the extra yap. That said, I wouldn't recommend using these as quick revision notes, but think of them as detailed notes covering almost everything

Spotted a mistake? Suggestions?
I'd love to hear your feedback. Contact us at: learnmates.share@gmail.com

Introduction

Types of reactions:

1. Addition

What happens: Two reactants combine to form a single product. No atoms are lost.

Key clue: The reactants are usually unsaturated (have a double/triple bond).

Example: Alkenes reacting with hydrogen (H_2), bromine (Br_2), or steam (H_2O).

$-A + B \rightarrow C-$

2. Elimination

What happens: A small molecule (like H_2O or HCl) is removed (eliminated) from a larger molecule, often forming a double bond.

Example: Heating a halogenoalkane or alcohol with NaOH/OH^- (or conc. H_2SO_4/H_3PO_4 for alcohols) to form an alkene.

3. Substitution

What happens: An atom (or group) is swapped for a different atom (or group).

Example: Halogenoalkane + NaOH (aqueous) → Alcohol. (Halogen swapped for OH).

$-A + B \rightarrow C + D-$

4. Oxidation

What happens: Gain of oxygen, loss of hydrogen, or loss of electrons. (Usually increases the oxygen/halogen content).

Example: Primary alcohol → Aldehyde → Carboxylic acid.

5. Reduction (a type of addition reaction)

What happens: Gain of hydrogen, loss of oxygen, or gain of electrons. (Opposite of oxidation).

Example: Alkenes to alkanes (using H_2/Ni catalyst) or Nitriles to amines (using $LiAlH_4$ or H_2/Ni).

6. Hydrolysis

What happens: A reaction with water (or aqueous OH^-) that breaks a chemical bond.

Example: Halogenoalkanes + aqueous NaOH (or water/$AgNO_3$) → Alcohols.

7. Polymerisation

What happens: Many small molecules (monomers) join together to form a giant chain (polymer).

Example: Addition polymerisation of alkenes (e.g., ethene → poly(ethene)).

Free Radical Substitution

Definition

A reaction mechanism involving free radicals (species with unpaired electrons) where bonds are broken by homolytic fission.

General Equation

Definition: Free Radical

A species with an unpaired electron.

Bond Breaking: Homolytic Fission

Each atom takes one electron from the covalent bond.

Shown using half-headed curly arrows (fishhooks).

Products are free radicals.

Electrophilic substitution

Definition

A reaction mechanism in which an electrophile attacks a double bond, causing it to open and add two atoms/groups to the carbons.

General Equation

Definition: Electrophile

An electron pair acceptor. An ion or molecule that accepts a pair of electrons to form a new covalent bond.

Bond Breaking: Heterolytic Fission

Both electrons go to one atom.

Products are ions (positive and negative species).

Leads to electrophiles and nucleophiles.

Nucleophilic Substitution

Definition

A reaction mechanism in which a nucleophile donates a lone pair to a electron-deficient carbon, swapping the halogen for another atom/group.

General Equation

Definition: Nucleophile

An electron pair donor. An ion or molecule that donates a lone pair of electrons to form a new covalent bond (nucleus loving)

Bond Breaking: Heterolytic Fission

Both electrons go to one atom (the halogen).

Products are ions.

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link between Bond Polarity and Mechanism:

Polar bonds (e.g., C-X in halogenoalkanes) undergo heterolytic fission. The electrons are pulled towards the more electronegative atom. This creates ions ($ \delta+$ and $\delta-$) and leads to ionic mechanisms (electrophilic/nucleophilic).-

Non-polar bonds (e.g., Cl-Cl in alkanes) can undergo homolytic fission (usually under UV light). Electrons are shared equally, so each atom takes one. This creates free radicals and leads to radical mechanisms.

Halogenoalkanes

Intro to halogenalkanes

Halogenoalkanes are alkanes where one or more hydrogen atoms have been replaced by a halogen atom (F, Cl, Br, I).

General formula: $C_nH_{2n+1}X$

Nomenclature

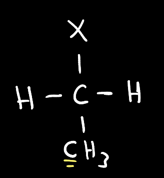
The naming stystem has already been explained in previous notes (alkanes U1) but here are some exmaples:

Name	Formula	How to name it
Chloromethane	CH_3Cl	Chloro + methane
Bromoethane	C_2H_5Br	Bromo + ethane
1–Chloropropane	$CH_3CH_2CH_2Cl$	Chloro on C–1 of propane
2–Bromopropane	$CH_3CHBrCH_3$	Bromo on C–2 of propane

Classification

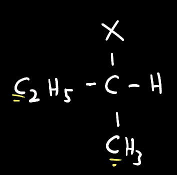
Halogenoalkanes can be classified as primary, secondary or tertiary depending on the number of carbon atoms attached to the C–X functional group

Primary (1°)



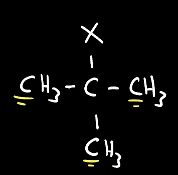
The carbon attached to the halogen is bonded to 1 carbon atom and 2 hydrogens.

Secondary (2°)



The carbon attached to the halogen is bonded to 2 carbon atom and 1 hydrogens.

Tertiary (3°)



The carbon attached to the halogen is bonded to 3 carbon atom and 0 hydrogens.

Reactions

Unlike alkane halogenalkanes are reactive molecules because they have Polar Bond: The C–X bond is polar (C are δ⁺ while X is δ⁻). The carbon is electron–deficient and wants to be attacked by electron–rich species.

Depending on the conditions (reagent and solvent), halogenoalkanes can react in two different ways:

Pathway	What happens	Conditions
Nucleophilic Substitution	The halogen is **swapped** for another atom/group.	**Aqueous** conditions (OH^- in water) or with nucleophiles like CN^-, NH_3.
Elimination	The halogen and a hydrogen are **removed**, forming a **double bond** (alkene).	**Ethanolic** conditions (OH^- in ethanol) with **heat**.

The Role of the Hydroxide Ion

The hydroxide ion (OH^-) is special because it can act in two different ways:

- As a Nucleophile: Donates a lone pair to form a new bond. Happens in aqueous conditions. Product = Alcohol
- As a Base: Accepts a proton (H^-) from the carbon next to the halogen. Happens in ethanolic conditions. Product = Alkene

Key Point: The solvent (water vs ethanol) decides the role of OH^-:
Aqueous: Substitution
Alcoholic: Elimination
(more on that later)

Recall:
Nucleophiles are electron pair donors (nucleus loving)

Reaction with <u>Aqueous</u> Potassium Hydroxide (KOH)

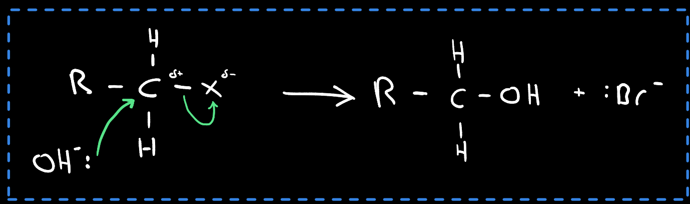
- Reagent: Potassium hydroxide dissolved in water (aqueous).
- Conditions: Heat under reflux.
- Product: Alcohol
- Role of OH^- (reagent): Nucleophile [Mechanism type: Nucleophilic substitution]

$$-R-X + OH^-(aq) \rightarrow R-OH + X^- -$$

The carbon attached to the halogen is δ⁺ (due to electronegative halogen). The hydroxide ion (OH^-) is electron–rich (negative charge and lone pairs). Opposites attract.

Mechanism: Primary halogenoalkane reacts with aqueous alkali

Bond Forms: lone pair on OH^- is donated to the δ⁺ carbon forming new covalent bond
Bond Breaks: As the new bond forms, the C–X bond breaks heterolytically and the electron pair leave with the bromine to form X^-
Products: Alcohol + anion X^-



Reaction with Alcoholic Ammonia (NH_3)

- Reagent: Ammonia dissolved in ethanol (alcoholic ammonia).
- Conditions: Heat excess ammonia in a sealed tube under pressure (1) in an alcoholic solution (2).
- Product: Amine (specifically a primary amine)
- Role of NH_3 (reagent): Nucleophile [Mechanism type: Nucleophilic substitution]

$$-R-X + NH_3 \rightarrow R-NH_2 + NH_4X-$$

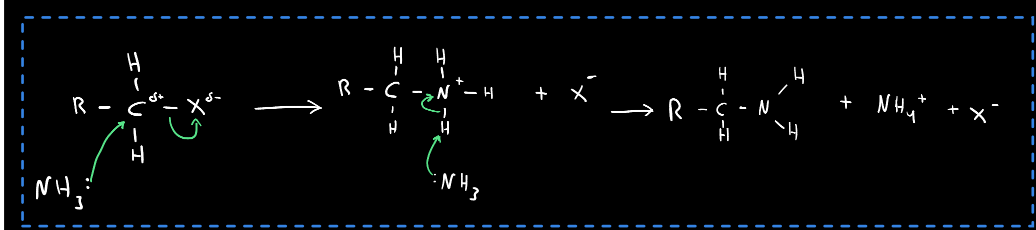
Why is a sealed tube used instead of reflux?
–If you try to heat a mixture of halogenoalkane and ammonia under reflux, the ammonia will boil away as a gas and escape through the condenser, therefor, a sealed tube is used since as you heat the tube, the ammonia vaporises but cannot escape.

Further Substitution: Why excess ammonia is essential?
–The Reactant: Ammonia ($:NH_3$) is a nucleophile. It attacks the halogenoalkane to form a primary amine ($R-NH_2$).
–The Product: The primary amine ($R-NH_2$) still has a lone pair on the nitrogen.
–The Consequence: The amine product can now act as a nucleophile and attack another halogenoalkane molecule (low yeild).

Step	Reaction	Product
1	$R\text{-}X + NH_3 \rightarrow$	**Primary amine** ($R\text{-}NH_2$) + HX
2	$R\text{-}X + R\text{-}NH_2 \rightarrow$	**Secondary amine** (R_2NH) + HX
3	$R\text{-}X + R_2NH \rightarrow$	**Tertiary amine** (R_3N) + HX
4	$R\text{-}X + R_3N \rightarrow$	**Quaternary ammonium salt** ($R_4N^+X^-$)

Mechanism: Primary halogenoalkane reacts with ammonia

Step 1: Nucleophilic Attack
Bond Forms: lone pair on NH_3 is donated to the δ⁺ carbon forming new covalent bond
Bond Breaks: As the new bond forms, the C–X bond breaks heterolytically and the electron pair leave with the bromine to form X^-
Intermediate: alkylammonium ion
Step 2: Proton Transfer
Bond Forms: Lone pair on a second ammonia molecule ($:NH_3$) donates to a proton (H^+) on the alkylammonium ion.
Bond Breaks: The N–H bond in the alkylammonium ion breaks heterolytically, with the electron pair staying on the nitrogen.
Products: Neutral primary amine ($CH_3CH_2NH_2$) + Ammonium ion (NH_4^+)



Halogenoalkanes

Reaction with Alcoholic Potassium Cyanide (KCN)

- Reagent: Potassium cyanide dissolved in ethanol (alcoholic)
- Conditions: Heat under reflux.
- Product: Nitrile
- Role of CN^- (reagent): Nucleophile [Mechanism type: Nucleophilic substitution]

$$R{-}X + CN^- \rightarrow R{-}CN + X^-$$

Example:

$$CH_3CH_2Br + KCN \rightarrow CH_3CH_2CN + KBr$$

bromoethane → propanenitrile

Notice that a carbon was added to the chain; the nitrile (−CN) introduces an extra carbon atom. This is useful for increasing carbon chain length in organic synthesis.

The carbon attached to the halogen is δ^+ (due to electronegative halogen). The cyanide ion (CN^-) is electron-rich (negative charge and lone pairs). Opposites attract.

Reaction with Ethanolic Potassium Hydroxide

- Reagent: Potassium hydroxide dissolved in ethanol
- Conditions: Heat under reflux.
- Product: Alkene
- Role of OH^- (reagent): Base (accepts a proton, does not attack the carbon) [Mechanism type: Elimination]

$$R{-}CH_2CH_2X + OH^-_{(ethanolic)} \rightarrow R{-}CH{=}CH_2 + X^- + H_2O$$

Example:

1— $CH_3CH_2Br + KOH \rightarrow CH{=}CH_2 + KBr + H_2O$

bromoethane → ethene

2— $CH_3CHBrCH_3 + KON \rightarrow CH_3CH{=}CH_2 + KBr + H_2O$

2-bromopropane → but-2-ene

In this reaction, the OH^- acts as a base accepting a proton (H^+) from the carbon adjacent to the one bearing the halogen. The C–X bond breaks heterolytically forming X^-, and a C=C double bond forms.

Note: If the halogenalkane is unsymmetrical (the two carbons adjacent to the C–OH are different), there are two possible hydrogens that can be removed. This leads to two different alkene products.

> The solvent matters!
> Aqueous KOH → substitution (alcohol);
> Ethanolic KOH → elimination (alkene)

Reaction with water (Hydrolysis)

- Reagent: Water (H_2O)
- Product: Alcohol
- Role of H_2O (reagent): Nucleophile [Mechanism type: Nucleophilic substitution]

$$R{-}X + H_2O \rightarrow R{-}OH + H^+ + X^-$$

Example:

$$CH_3CH_2Br + H_2O \rightarrow CH_3CH_2OH + HBr$$

Bromoethane → Ethanol

This reaction is slow because water is a weak nucleophile (unlike OH^-).

Compare Rates of Hydrolysis

Using Aqueous Silver Nitrate in Ethanol

<u>What is the reagent?</u>

Water (H_2O): The actual nucleophile that reacts with the halogenoalkane.

Ethanol (C_2H_5OH): The co-solvent that helps dissolve the mixture and makes it homogeneos

Halogenoalkanes are non-polar and would not mix well with water alone).

Silver nitrate ($AgNO_3$): The hydrolysis reaction releases halide ions (X^-), but we cannot see them. Adding silver nitrate ($AgNO_3$) causes a silver halide precipitate (AgX) to form. The time taken for the precipitate to appear indicates the rate of hydrolysis — faster precipitate = faster hydrolysis.

Comparison (i): Primary, Secondary and Tertiary halogenaalkanes

Aim: To compare the rate of hydrolysis of halogenoalkanes with the same halogen but different structures.

Method: Add aqueous silver nitrate in ethanol to each isomer at room temperature. Observe the time taken for a precipitate to form.

Type	Example	Relative Rate	Observation
Tertiary	$(CH_3)_3CBr$	Fastest	Precipitate forms **immediately** at room temperature
Secondary	$CH_3CHBrCH_3$	Medium	Precipitate forms within **minutes**
Primary	$CH_3CH_2CH_2Br$	Slowest	Precipitate forms **slowly**; may require **warming**

Order of Reactivity: Tertiary > Secondary > Primary

Comparison (ii): Chloro-, Bromo- and Iodoalkanes

Aim: To compare the rate of hydrolysis of halogenoalkanes with different halogens.

Method: Add aqueous silver nitrate in ethanol to each isomer at room temperature. Observe the time taken for a precipitate to form.

Halogenoalkane	Bond	Bond Strength (kJ/mol)	Relative Rate	Observation
Iodoalkane	C–I	238	Fastest	Precipitate forms quickly at room temperature
Bromoalkane	C–Br	276	Medium	Precipitate forms slower than iodo
Chloroalkane	C–Cl	338	Slowest	Precipitate forms very slowly; may require warming

Order of Reactivity: Iodo > Bromo > Chloro

Explanation:

As you go down Group 7, atomic size increases → bond length increases → bond strength decreases.

The weaker the C–X bond (lower bond enthalpy) → the easier the halogen leaves, so the faster the hydrolysis.

C–I is the weakest bond → fastest reaction.

C–Cl is the strongest bond → slowest reaction.

Bond Polarity ≠ Reactivity

Fair testing

When comparing rates of hydrolysis, all other factors must be kept constant to ensure a fair comparison:

- Concentration: Same concentration of halogenoalkane and silver nitrate ensures rate differences are due to structure or halogen, not concentration.
- Temperature: All reactions should be at the same temperature (e.g. room temperature) because higher temperature increases rate regardless of the halogenoalkane.
- Solvent composition: Same ratio of water to ethanol ensures the solvent environment is identical for each reaction.
- Volume of solution: Same total volume ensures consistent mixing and observation.
- Same halogen type [For comparison (i)]: e.g. all are bromoalkanes
- Same structure [for comparison (ii)]: e.g. all are primary halogenalkanes

Why this matters: By keeping these factors constant, any difference in the time taken for precipitate to form can be attributed only to:

1)The structure of the halogenoalkane (primary/secondary/tertiary)

OR

2)The halogen present (Cl/Br/I)

Alcohols

Intro to alcohols

Alcohols are organic compounds containing a hydroxyl group (–OH) bonded to a saturated carbon atom.

General Formula: $C_nH_{2n+1}OH$ (or $C_nH_{2n+2}O$)

Nomenclature

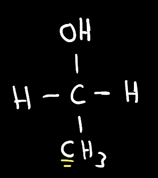
The naming stystem has already been explained in previous notes (alkanes U1) but here are some exmaples:

Name	Formula	How to name it
Methanol	CH_3OH	methan + ol
Ethanol	C_2H_5OH	ethan + ol
Propan–1–ol	$CH_3CH_2CH_2OH$	propan + ol (OH on carbon 1)
Propan–2–ol	$CH_3CH(OH)CH_3$	propan + ol (OH on carbon 2)

Classification

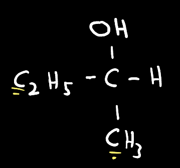
Alcohols can be classified as primary, secondary or tertiary depending on the number of carbon atoms attached to the carbon brearing –OH group.

Primary (1°)



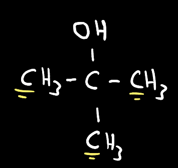
The carbon attached to the –OH is bonded to 1 carbon atom and 2 hydrogens.

Secondary (2°)



The carbon attached to the –OH is bonded to 2 carbon atom and 1 hydrogens.

Tertiary (3°)



The carbon attached to the –OH is bonded to 3 carbon atom and 0 hydrogens.

Alcohols: Reactions

Combustion (Reaction with Oxygen in Air)

Alcohols undergo complete combustion in excess oxygen to produce carbon dioxide and water. This reaction is exothermic (releases heat).

General Equation:

—Alcohol + Oxygen → Carbon dioxide + Water—

Example (Ethanol):

$$C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$$

Observations:

Burns with a clean, blue flame (similar to alkanes)

Produces heat energy — used as a fuel (e.g., ethanol in spirit burners, biofuel)

Dehydration (Elimination) with Concentrated Phosphoric Acid

Alcohols undergo elimination (dehydration) to form alkenes. A water molecule is removed from the alcohol, creating a C=C double bond.

Reagent: Concentrated phosphoric acid (H_3PO_4) — acts as a catalyst and dehydrating agent

Conditions: Heat

General Equation: $C_nH_{2n+1}OH \xrightarrow{H_3PO_4, \Delta} C_nH_{2n} + H_2O$

During dehydration, the –OH group and a hydrogen atom from an adjacent carbon are removed as water. A C=C double bond forms between the two carbons.

If the alcohol is unsymmetrical (the two carbons adjacent to the C–OH are different), there are two possible hydrogens that can be removed. This leads to two different alkene products.

> Alternative: Al_2O_3 can also be used as a dehydrating agent. The alcohol is passed as a vapour over hot Al_2O_3 (300–350°C) to form the alkene.

Halogenating Agents

Alcohols can be converted into halogenoalkanes by replacing the –OH group with a halogen atom (Cl, Br, or I).

Producing Chloroalkanes

Reagent: Phosphorus(V) chloride (PCl_5)

Product: Chloroalkane

Genaral Equation:

$$R\text{–}OH + PCl_5 \rightarrow R\text{–}Cl + POCl_3 + HCl$$

Observations:

- Steamy/misty white fumes of HCl are produced
- This is used as a qualitative test for the presence of the –OH group

> Chlorination of tertiary alcohols can be done in a different method (both methods work), that does not work with primary and secondary alcohols, and that done is by mixing (shaking) the alcohol with hydrochloric acid at room temperature.
> A common example is: $(CH_3)_3COH + HCl \rightarrow (CH_3)_3CCl + H_2O$

Producing Bromoalkanes

Reagent: Potassium bromide (KBr) + 50% concentrated sulfuric acid (H_2SO_4)

Product: Bromoalkane

<u>What happens:</u> The H_2SO_4 reacts with KBr to produce HBr *in situ* (within the reaction mixture) then HBr reacts with the alcohol to form the bromoalkane.

<u>Important:</u> 50% concentrated sulfuric acid is used as more concentrated sulfuric acid would oxidise bromide ions to bromine (Br_2), resulting in different products (e.g. bromine instead of HBr, leading to possible side reactions).

Step 1: Formation of HBr:

$$KBr + H_2SO_4 \rightarrow HBr + KHSO_4 \text{ or } 2KBr + H_2SO_4 \rightarrow 2HBr + K_2SO_4$$

Step 2: Substitution with Alcohol:

$$R\text{–}OH + HBr \rightarrow R\text{–}Br + H_2O$$

In situ (latin: "in place"):

A reagent is produced within the reaction mixture and reacts immediately, rather than being added as a separate reagent.

Alcohols

Producing Iodoalkanes

Reagent: Red phosphorus and iodine (I_2)
Product: Iodoalkane

What happens: Red phosphorus and iodine react together to form phosphorus(III) iodide (PI_3) *in situ* (within the reaction mixture), which then reacts with the alcohol.

Step 1: Formation of PI_3:

$$2P + 3I_2 \rightarrow 2PI_3$$

Step 2: Substitution with Alcohol:

$$3R{-}OH + 2PI_3 \rightarrow 3R{-}I + H_3PO_3$$

Oxidation

Reagent: acidified potassium dichromate(VI) ($K_2Cr_2O_7$ + dilute H_2SO_4)
Observation (if acohol is oxidized):
–Solution color changes from orange [dirchromate(VI) ion ($Cr_2O_7^{2-}$)] to green [Chromium(III) ions (Cr^{3+})]

What Happens During Oxidation?
Oxidation of alcohols involves the **removal of hydrogen** atoms from the molecule.

Specifically:
- The –OH group loses its hydrogen atom.
- The carbon atom bearing (not carbon the adjecent) the –OH group loses one hydrogen atom (if available)

These two hydrogens combine with oxygen from the oxidising agent to form water (H_2O).

General reaction: $Alcohol \xrightarrow{[O]} Aldehyde/Ketone + H_2O$

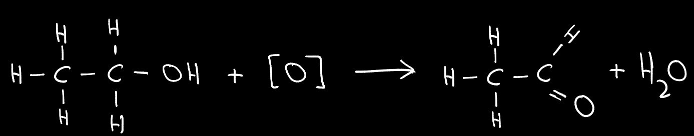
For oxidation to occur, the carbon bearing the –OH group must have at least one hydrogen atom attached to it.

Alcohol Type	Structure	Hydrogens on C–OH	Oxidation Possible?
Primary (1°)	R-CH_2OH	2 hydrogens	Yes — can be oxidised twice
Secondary (2°)	R_2CHOH	1 hydrogen	Yes — can be oxidised once
Tertiary (3°)	R_3COH	0 hydrogens	No — no hydrogen to remove

Primary Alcohols: Partial Oxidation

A primary alcohol loses two hydrogen atoms (one from the –OH group and one from the carbon) to form an aldehyde.
These aldehydes can vapidly get oxidized further to carboxylic acids → thats why to obtain aldehydes we use distillation with addition (explained in more details in later sections)

General equation: $R{-}CH_2OH + [O] \rightarrow R{-}CHO + H_2O$
Example:



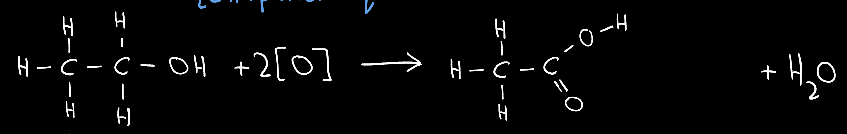
Ethanol Ethanal

Primary Alcohols: Full Oxidation

A primary alcohol is oxidised twice — first to an aldehyde, then further to a carboxylic acid.
To achieve this we carry out heating under reflux (explained in more details in later sections)

General equation: $R{-}CH_2OH + [O] \rightarrow R{-}CHO + H_2O$
then $R{-}CHO + [O] \rightarrow R{-}COOH$
Example:

[Simplified equation]

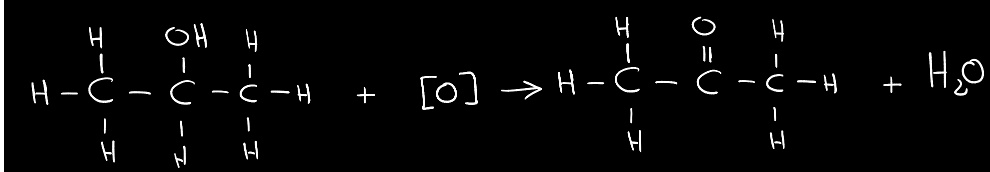


Ethanol Ethanoic Acid

Secondary Alcohols

A secondary alcohol loses two hydrogen atoms (one from the –OH group and one from the carbon) to form a ketone. Unlike primary alcohols, ketones cannot be oxidised further.

General equation: $R_2CHOH + [O] \rightarrow R_2CH{=}O + H_2O$
Example:



Propan-2-ol Propanone

Tests

Step	Action	Observation	Conclusion	Chemical Explanation
1	Add acidified $K_2Cr_2O_7$	Orange remains	**Tertiary alcohol**	No H on C–OH → cannot be oxidised
		Orange → Green	**Primary or secondary**	Oxidation occurred → $Cr_2O_7^{2-}$ reduced to Cr^{3+}
2	**Choose technique based on what you want to find out:**			
	Distillation with addition (collect product)	Product collected	—	Aldehyde or ketone is distilled off as formed
	Heat under reflux (full oxidation)	Product in flask	—	Aldehyde (if formed) remains and is further oxidised
3a	Add Benedict's/Fehling's to distilled product (heat gently)	**Brick-red precipitate**	**Aldehyde** → Primary alcohol	Aldehyde reduces Cu^{2+} to Cu_2O (brick-red)
		No change (blue remains)	**Ketone** → Secondary alcohol	Ketone cannot reduce Cu^{2+}
3b	Add Na_2CO_3 or $NaHCO_3$ to reflux mixture	**Effervescence** (CO_2 bubbles)	**Carboxylic acid** → Primary alcohol	Acid + carbonate → CO_2 gas
		No change	**Ketone** → Secondary alcohol	

Other tests

Tollens reagent

Reagent: Aqueous ammonia with silver nitrate
Detects: Aldehydes
Conditions: Warm gently
Observation: Silver mirror forms on test tube walls
Explanation: Aldehyde is oxidised to carboxylic acid; Ag^+ is reduced to $Ag_{(s)}$
Ketones: No reaction

Reaction with sodium

Reagent: Sodium metal Na
Detects: –OH group (alcohols, carboxylic acids...)
Conditions: Room temperature, dry conditions
Observation: Effervescence (bubbles of H_2 gas) + sodium disappears
Confirmation: Squeaky pop test with a lighted splint
Equation: $2R{-}OH + 2Na \rightarrow 2R{-}ONa + H_2$